



Tractor Rage: Intellectual Property, Agriculture, Competition Policy, and the Right to Repair

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Accepted: 21 October 2024 / Published online: 31 January 2025
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Abstract This article considers the relationship between intellectual property and competition policy in the context of agriculture in the United States. It provides an analysis of the debate over the right to repair in respect of agricultural machinery. It offers a case study of John Deere – the American company, specializing in agricultural machinery, which has been at the centre of many debates over the right to repair. This article considers the consolidated class action by farmers against John Deere. It explores efforts to establish state right to repair laws focusing on agricultural machinery – highlighting the pioneering example of Colorado as a model to be followed elsewhere. This paper considers the debate over Federal copyright laws and the application of technological protection measures to agricultural machinery. It examines various proposals in the United States Congress to recognise an agricultural right to repair. It explores the impact of President Joe Biden’s executive order, and the enforcement action of the Federal Trade Commission on the right to repair. The conclusion considers the opportunities for the reconfiguration of the relationship in respect of intellectual property and competition policy in the field of agriculture and farming. It contends that the recognition of a broad right to repair in agriculture will promote farmers’ rights, competition policy, and sustainable development in the United States.

Keywords Right to repair · Intellectual property · Competition policy · Agriculture

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1 Introduction

Hearing a class action by Illinois farmers against the Illinois-derived company John Deere over repair restrictions in 2023, the District Court judge Iain D. Johnston reflected upon the historical significance of the litigation:

Less than an hour's drive from the Stanley J. Roszkowski U.S. Courthouse down Illinois Route 2 between Oregon and Dixon, Illinois is the John Deere Historic Site. Free of charge, visitors can learn much about John Deere, the man. The takeaway from a visit to this historic site is that John Deere was an innovative farmer and blacksmith who – with his own hands – fundamentally changed the agricultural industry. This multi-district litigation concerns allegations of non-competitive behavior by Deere & Co., a multi-billion-dollar international corporation. If – and that's a big if – the claims against Deere & Co. are meritorious, then the Court assumes the man lionized at the historic site would be deeply disappointed in his namesake corporation.¹

The judge cautioned: “The ultimate determination of the claims will likely be a long and expensive process, despite this Court's goal of bringing this litigation to a just, speedy, and inexpensive resolution.”² The company John Deere has become a key focal for litigation and the broader public debate over intellectual property, competition policy, and the right to repair in the United States and other key jurisdictions.

John Deere is an American company, which specialises in the manufacturing of agricultural machinery, construction equipment, and forestry equipment. The company dates back to the 19th century in Illinois. John Deere relies upon a combination of forms of intellectual property – including trade mark law, patent law, copyright law, and trade secrets. The company has used the logo of a leaping deer as its trade mark, and has deployed the slogan, “Nothing Runs Like a Deere.” John Deere have built a substantial portfolio of patents. According to a recent patent landscape analysis, the company had a portfolio of 32,487 patents globally – of which 14,260 had been granted.³ Of those 32,487 patents, 20,778 patents were active.⁴ The company had 9,921 unique patent families.⁵ John Deere has been involved in a number of patent conflicts with its rivals and competitors.⁶ John Deere has also relied upon copyright protection in respect of computer programs and software associated with its machinery. Aaron Perzanowski comments: “Copyright law has been central to Deere's strategy to shut competitors out of the lucrative market for farm-equipment repair.”⁷ The company has lobbied against exceptions

¹ *In re Deere & Company Repair Service Antitrust Litigation* United States District Court, N.D. Illinois, Western Division. 27 November 2023, 2023 WL 8190256.

² *Ibid.*

³ Grey (2023).

⁴ *Ibid.*

⁵ *Ibid.*

⁶ Jones (2023), Brittain (2024).

⁷ Perzanowski (2022), p. 9.

for repairs being granted under copyright law and technological protection measures.⁸ The company has also sought to protect its confidential information and trade secrets as well – particularly in disputes with former employees.⁹

In terms of intellectual property public policy, John Deere has gained a reputation for being a key critic and opponent of the right to repair.¹⁰ The company – and the lobby group the Equipment Dealers Association – have opposed Federal and State right to repair initiatives in the United States.¹¹ The company has instead taken the position that the industry should self-regulate in respect of repairs.¹² John Deere was also a critic of the proposal for a right to repair in Australia.¹³ The company faced significant lawsuits from farmers over such repair restrictions. The company has also been at the centre of much of the policy debate around the right to repair in the field of intellectual property and agriculture. In response, John Deere has made concessions on its stance on the right to repair. The right to repair movement remains sceptical of John Deere's commitment to repair.¹⁴ While this paper focuses on John Deere as an exemplar, it is by no means not the only such company embroiled in the debate over the right to repair. Caterpillar, the manufacturer of construction and mining equipment, has been a vocal opponent of the right to repair.¹⁵ Electric car manufacturer Tesla has also battled litigation over the right to repair.¹⁶ The information technology company Apple has faced scrutiny over its approach to the right to repair.¹⁷

In terms of its methodology, this article engages in a corporate case study of John Deere, focusing on the interplay of intellectual property, competition policy, and the right to repair. This is an original study. While John Deere has featured as a player in academic literature on the right to repair, there has been no sustained study of the company itself. This study is not mere reportage or journalism – it is an effort to provide a consolidated understanding of the dynamic role of the company in litigation and policy reform. In the process, it unearths a range of original material, including: emerging litigation, pioneering law reform, and regulatory responses to new technologies and business models. Such a corporate case study approach has been successfully deployed by technology scholars, particularly in the United States, looking at parallel scenarios. In the field of biotechnology, Sally Smith Hughes has used a study of Genentech to shed light on the formation of bioethics.¹⁸ Jorge Contreras has studied gene patents, through an in-depth analysis of Myriad

⁸ Forelle (2023).

⁹ *Deere & Company v. Crawford et al*, 4:2019cv04178 (US District Court for the Central District of Illinois, 9 September 2019).

¹⁰ Kyle Wiens of iFixit has been a key critic of John Deere. Wiens (2015).

¹¹ Geangan (2022).

¹² Patel (2021).

¹³ John Deere Limited (2021).

¹⁴ Hernandez (2023), O'Reilly (2023).

¹⁵ Bushey (2019), Sessa-Hawkins (2017).

¹⁶ Scarcella (2023a).

¹⁷ Rognstad (2021), Labib (2023).

¹⁸ Hughes (2011).

Genetics.¹⁹ Patrick Radden Keefe offered an analysis of the opioid crisis through a case study of Purdue Pharma.²⁰ John Carreyrou highlighted issues of fraud in intellectual property and health regulation in a study of Theranos.²¹ In the field of agriculture, Bartow Elmore has conducted a history of the Monsanto chemical and agribusiness.²² By deploying a corporate history methodology, it is hoped that this study sheds light on John Deere, and the larger issues in respect of intellectual property and the right of repair in the field of agriculture and farming.

This article contends that the recognition of a broader right to repair in agriculture will boost farmers' rights, competition, and sustainable development. It nonetheless highlights the challenges involved in achieving law reform in respect of the right to repair in the United States. The topic of the right to repair cuts across state law, Federal law, and executive orders in the United States system. Moreover, the right to repair raises a cross-disciplinary array of legal issues – including contract law, consumer rights, competition policy, intellectual property law, environmental law, and data protection. There is a need to develop an integrated, and holistic approach to the right to repair in the United States. Lawrence Lessig has highlighted how intellectual property reforms can be delayed, blocked, and thwarted through political lobbying, interference, and corporate capture.²³ He has emphasized that there is a need to analyse the process of law reform, and not just the outcomes of policy debates.²⁴ His work has showcased how there is a need to take into account laws, social norms, market forces, and technological innovation. Such an understanding of political science informs this analysis of the debate over the right to repair in the United States.

This article provides a comprehensive analysis of the right to repair in the field of farming – focusing on conflicts over agricultural machinery, particularly involving the company John Deere in the United States. Section 2 considers the consolidated class action by farmers against John Deere in the United States, alleging breaches of the Sherman Act. Section 3 focuses on the Memorandum of Understanding between the American Farm Bureau Federation and John Deere. Section 4 explores the efforts to establish an agricultural right to repair at a state level – highlighting the pioneering example of Colorado. Section 5 explores the federal debate over copyright law, technological protection measures, and the repair of agricultural machinery. Section 6 explores various efforts in the United States Congress to recognise an agricultural right to repair – most notably, Senator Jon Tester's *Agricultural Right to Repair Act*. Section 7 looks at President Joe Biden's executive order on competition, and its application by the Federal Trade Commission in respect of the right to repair. The conclusion draws some broader recommendations about the need a further realignment of intellectual property and competition policy – focusing on agriculture and farming.

¹⁹ Contreras (2021).

²⁰ Keefe (2021).

²¹ Carreyrou (2018).

²² Elmore (2021).

²³ Lessig (1999, 2001, 2004, 2008).

²⁴ Lessig (2015, 2018, 2019).

2 Consolidated Class Action

In the field of agriculture, there has been an increasing concern about the rise of monopolies and oligopolies. In this context, there has been a particular grievance about the lack of a right to repair in respect of tractors and other agricultural machinery.

2.1 Literature Review

A series of books have highlighted competition concerns in respect of agricultural machinery in the United States.

In her book, *Break 'em Up*, Zephyr Teachout highlights the dominant market power of the company John Deere:

Even on his tractor, a farmer has no freedom: John Deere contracts require that the tractors be serviced by affiliates of the companies. In other words, you're not allowed to fix your own tractor. And John Deere's power is built on over 20 mergers, and the fact that it has made itself the fifth-largest agricultural lender in the country. It can issue credit to farmers on the condition that they buy tractors from Deere and not the competition. A third of Deere's profits are now in lending, and it leverages its dominance in each arena to increase its dominance in the other. John Deere's profits are up, farmers' are down; during a recent downturn, when farmers struggled, Deere was doing just fine.²⁵

Zephyr Teachout commented that the agricultural and farming markets are dominated by a "few corporations, depriving farmers of security, dignity, and the freedom to be full citizens."²⁶

In his book, *Monopolized*, David Doyen is also worried about the corporate concentration of ownership in respect of tractors and agricultural machinery: "Two companies, Deere & Co. and Dutch firm CNH Industrial, sell about half of all tractors and other farm machinery, and they prohibit farmers from fixing them by restricting access to the embedded software needed to facilitate repairs."²⁷ He noted that this system "funnels thousands of dollars in service costs to manufacturers and their authorized agents every time there's even a minor breakdown."²⁸ Dayen commented: "You could miss a harvest and lose your farm just because you can't fix a locked machinery circuit."²⁹ He observed: "An entire network of underground hacking and a 'right to repair' movement have emerged to fight the tractor monopoly."³⁰ Dayen observed: "The industry has spent millions to block both, with

²⁵ Teachout (2020), p. 26.

²⁶ *Ibid.*, p. 27.

²⁷ Dayen (2020), p. 55.

²⁸ *Ibid.*, p. 55.

²⁹ *Ibid.*, p. 55.

³⁰ *Ibid.*, p. 55.

Deere even asserting that when farmers buy a tractor, they merely own a ‘license to operate the vehicle’.”³¹

Aaron Perzanowski comments: “In the market for farm equipment, where companies like John Deere have taken aggressive steps to limit competition, repair is five times as profitable as equipment sales.”³² He questions the approach of John Deere to product design: “When John Deere designs its tractors so that the authentic replacement parts won’t work without its initialization software, that is a predatory decision.”³³ He observes that such an approach offers “little benefit to equipment owners, but imposes significant barriers for repair providers.”³⁴ There has been much concern amongst farmers as to John Deere’s product design.³⁵ Perzanowski is sceptical of the argument of John Deere that the recognition of the right to repair will undermine environmental standards on emissions: “John Deere and its friends in the heavy equipment industry have tried to convince lawmakers that recognizing farmers’ right to repair their tractors will open the door to ‘tampering’ and violations of the federal emissions standards.”³⁶

2.2 Class Actions

There has been a wave of class actions launched by farmers in various parts of the United States against John Deere. In January 2022, a Forest River, North Dakota farm owned by Robert Blair sued John Deere in a federal class action suit in the United States District Court for the Northern District of Illinois for the Eastern Division.³⁷ The case of *Trinity Dale Wells v. John Deere* involves an antitrust class action led by Trinity Dale Wells in the United States District Court for the Northern District of Alabama.³⁸ A group of corporate and individual farmers led by Plum Ridge Farms Ltd brought a putative multidistrict (MDL) class action alleging that manufacturer of tractors and other agricultural machinery acted in concert with its authorised dealerships to withhold software tools and resources required to repair the tractors from competitors, in violation of the *Sherman Act*.³⁹ David Underwood led a class action complaint against John Deere in the State of Tennessee.⁴⁰ Monty Ferrell led a class action complaint against John Deere in the state of

³¹ *Ibid.*, p. 55.

³² Perzanowski (2022), p. 23.

³³ *Ibid.*, p. 189.

³⁴ *Ibid.*, p. 189.

³⁵ Roberts (2023).

³⁶ Perzanowski (2022), p. 233.

³⁷ *Forest River Farms v. John Deere & Co.* 1: 22-cv-00188; and Pates (2022).

³⁸ *Trinity Dale Wells v. John Deere Case* (2022) U.S. District Court for the Northern District of Alabama, 3:22-cv-00074-LCB

³⁹ Class Action Complaint in *Plum Ridge Farms Ltd.*, individually and on behalf of all others similarly situated, Plaintiff, v. Deere & Co. (d/b/a John Deere), Defendant. United States District Court, N.D. Illinois 2022 WL 2914529.

⁴⁰ Class Action Complaint. *Underwood v. Deere & Co.* United States District Court, E.D. Tennessee 2022 WL 340660.

Oklahoma.⁴¹ Virginia farmer Samantha Casselbury brought a class action against John Deere in the United States District Court for the Central District of Illinois in the Rock Island Division.⁴² This class action litigation has been consolidated in a dispute in the United States District Court for the Northern District of Illinois (Western Division) under the supervision of the judge, the Hon. Iain D. Johnston. It is worthwhile considering the consolidated class action compliant of the various farmers.⁴³

2.3 Complaint

In its introduction, the complaint explains:

This case is about John Deere and its affiliated dealerships' monopolization, attempted monopolization, exclusionary conduct, and restraint of the repair service market for John Deere ("Deere") brand agricultural equipment (such as tractors, combines, and other large agricultural machinery) with onboard computers known as electronic control units, or "ECUs".⁴⁴

The litigation involves an antitrust class action under Secs. 1 and 2 of the Sherman Act brought on behalf of farmers against Deere and Co and Deere dealerships.

The lead plaintiffs include Plum Ridge Farms Ltd., an agricultural crop farm in Elizbeth, Illinois; Colvin Farms Ltd, a farm in Marianna, Florida; England Farms & Harvesting LLC from Mercedes, Texas; Robbins Family Grain Co. LLC from Sackets Harbor, New York; Wilson Farms Land & Cattle Co Ltd, LL from Butler, Missouri; Hapka Farms Inc. from Warren, Minnesota; Eagle Lake Farms Partnership from Newport, Arkansas; Blake Johnson of Richie, Tennessee; and Trinity Dale Wells, a resident of Spruce Pine, Alabama.

The complaint makes a number of factual allegations. In essence, the class action maintains: "Farmers historically have been able to perform their own repairs on their Tractors or have them repaired by their independent mechanic of choice."⁴⁵ Nonetheless, the complaint observes: "However, as computers and software have become increasingly intertwined with basic operations of farming equipment, Deere has restricted access to necessary Repair Tools, thereby preventing owners and independent repair shops from competing with Deere-affiliated Dealerships to make repairs on newer, ECU-equipped Tractors."⁴⁶

⁴¹ Class Action Complaint. *Ferrell v. Deere & Co.* United States District Court, W.D. Oklahoma. 2022 WL 860145.

⁴² Class Action Complaint Jury Trial Demanded in *Casselbury v. John Deere* United States District Court, C.D. Illinois 2022 WL 3368271.

⁴³ Consolidated Class Action Complaint, *In Re: Deere & Company Repair Services Antitrust Litigation* Case 3:22-cv-50188 (US District Court for the Northern District of Illinois, Western Division), 24 October 2022.

⁴⁴ *Ibid.*, p. 3.

⁴⁵ *Ibid.*, p. 20.

⁴⁶ *Ibid.*, p. 20.

In terms of the class action allegations, the farmers argued that their claims “are typical of the claims of the members of the Class because Plaintiffs purchased Deere Repair Services from the co-conspirator Dealerships, and therefore Plaintiffs’ claims arise from the same common course of conduct giving rise to the claims of the Class and the relief sought is common to the Class.”⁴⁷ The farmers argued that a class action was superior, because “individual litigation presents the potential for inconsistent or contradictory judgments and would greatly magnify the delay and expense to all parties and to the court system.”⁴⁸

The class action alleged that the anticompetitive conduct of Deere and its co-conspirators had caused various forms of antitrust injury. The class action argued: “The purpose of Deere and its co-conspirators’ conduct was to exclude competition and raise, fix, or maintain the price of Deere Repair Services.”⁴⁹ The class action maintained: “As a direct and foreseeable result, Plaintiffs and the Class paid supracompetitive prices for Deere Repair Services during the Class Period.”⁵⁰

There were various claims of relief. Count One was for a violation of Sec. 1 of the Sherman Act through a conspiracy in restraint of trade: “Defendant and co-conspirator Dealerships engaged in anticompetitive behavior by agreeing to withhold necessary Repair Tools from farmers and independent repair shops, which unfairly suppressed price competition for Deere Repair Services and unreasonably restrained trade.”⁵¹

Count Two was for a violation of Sec. 1 of the Sherman Act via a group boycott: “Defendant’s and the Co-conspirator Dealerships’ boycott cut off access to the Repair Tools, a necessary resource that would enable farmers and independent repair shops to compete in the Deere Repair Services market.”⁵²

Count Three was for a violation of Sec. 1 of the Sherman Act through an unlawful tying arrangement: “Plaintiffs and Class members through their purchase of Deere Tractors were coerced into purchasing a second tied product, Deere Repair Services, from Defendant and its Co-Conspirator Dealerships.”⁵³

Count Four was for a violation of Sec. 2 of the Sherman Act through monopolization: “As a direct and proximate result of Deere’s anticompetitive and monopolistic conduct, Plaintiffs and the Class have been damaged by, among other things: (i) the payment of supracompetitive prices for Deere Repair Services; and (ii) the lack of availability of independent Deere Repair Services and self-repair options.”⁵⁴

Count Five was for a violation of Sec. 2 of the Sherman Act through monopoly leveraging. It was alleged that “Deere has willfully and intentionally used its

⁴⁷ *Ibid.*, p. 57.

⁴⁸ *Ibid.*, p. 59.

⁴⁹ *Ibid.*, p. 60.

⁵⁰ *Ibid.*, p. 60.

⁵¹ *Ibid.*, p. 60.

⁵² *Ibid.*, p. 62.

⁵³ *Ibid.*, p. 63.

⁵⁴ *Ibid.*, p. 65.

monopoly power over Deere Repair Tools to gain or attempt to gain or maintain monopoly power in the Deere Repair Services Market.”⁵⁵

Count Six was for a violation of Sec. 2 of the Sherman Act through attempted monopolization in the alternative: “Deere’s anticompetitive conduct alleged herein has been directed at accomplishing the unlawful objective of controlling prices and/or preventing competition in the Repair Services Market.”⁵⁶

Count Seven was for a violation of Sec. 2 of the Sherman Act through a conspiracy to monopolize: “Defendant’s and the Co-conspirator Dealerships’ boycott cut off access to the Repair Tools, necessary resources that would enable farmers and independent repair shops to compete in the Deere Repair Services market.”⁵⁷

The plaintiffs sought declaratory and injunctive relief under the federal antitrust laws. There was a request for relief and a plea for a jury trial.

2.4 Preliminary Hearing

In a preliminary hearing, District Court judge Iain D. Johnston, reflected upon the significance of the litigation (as was discussed at the opening of this paper).⁵⁸ The judge summarized the complaint:

When a Tractor malfunctions, the farmer is effectively left to the “mercy” of Deere and the Dealerships. Farmers are “prevented from using trusted, less expensive, and more conveniently-located skilled mechanics who are not affiliated with Deere.” Independent mechanics “cannot effectively work on modern equipment due to [the] lack of access to the Repair Tools.” To make matters worse, Repair Services are both untimely and expensive. Dealerships could be “far away” and are frequently “understaffed.” Farmers “may have equipment sitting at the Dealership for months untouched by any technicians or may be waiting for days or weeks to have a technician travel to their farm.” Farmers are also responsible for not only the cost of new parts and Repair Services, but also for the cost of the technician’s travel Not surprisingly, Deere’s equipment is not interchangeable with its competitors.⁵⁹

The judge noted: “According to the complaint’s allegations, Deere has the ultimate control of the repair services market.”⁶⁰ The judge observed: “These allegations are not mere legal conclusions.”⁶¹ In his opinion, “The complaint is chock-full of factual allegations to support this conclusion.”⁶²

⁵⁵ *Ibid.*, p. 66.

⁵⁶ *Ibid.*, p. 67.

⁵⁷ *Ibid.*, p. 68.

⁵⁸ *In re Deere & Company Repair Service Antitrust Litigation* United States District Court, N.D. Illinois, Western Division. 27 November 2023, 2023 WL 8190256.

⁵⁹ *Ibid.*

⁶⁰ *Ibid.*

⁶¹ *Ibid.*

⁶² *Ibid.*

The judge held that the “Plaintiffs have sufficiently and plausibly alleged claims based on § 1 and § 2 of the *Sherman Act* under *Kodak*.”⁶³ The judge concluded that the “Plaintiffs’ Complaint alleges both constitutional and antitrust standing, relevant markets, and all the necessary requirements for each count in the Complaint.”⁶⁴ The judge observed: “Deere’s motion for judgment on the pleadings is denied.”⁶⁵ The preliminary decision received broad media attention.⁶⁶ A lead plaintiffs’ attorney in the case, Daniel Hedlund of law firm Gustafson Gluek in Minneapolis, commented that they “look forward to continuing to pursue the claims of their farmer clients in this right to repair antitrust case.”⁶⁷

The United States Department of Justice has issued a Statement of Interest in the ongoing antitrust litigation versus John Deere.⁶⁸ The submission highlights the importance of the right to repair, citing: “There is a growing body of economic literature and consumer effort to protect consumers’ freedom to repair their own products.”⁶⁹ The United States Department of Justice expressed concern that repair restrictions on agricultural machinery could have adverse impacts upon American farmers:

The repair restrictions at issue here affect Deere agricultural equipment that are important, costly investments to the workings of a farm. These various machines, or “tractors” for short, enable American agriculture. When they break or fail to operate and repair markets function poorly, agriculture suffers. Crops waste. Land lies fallow. Even a short delay can result in farmers “watch[ing] their crops rot.”⁷⁰

The submission is highly critical of John Deere’s reading of the law, declaring that “Deere is wrong.”⁷¹

At the 2023 Food and Agribusiness National Conference, Michael Kades – a Deputy Assistant Attorney General at the U.S. Department of Justice – discussed competition law and policy in the context of agricultural markets.⁷² He highlighted the importance of the topic of the right to repair. Kades discussed the conflict in respect of John Deere’s tractors: “Deere asserts that, unless it surprises or deceives customers, its limitations on repair rights cannot violate the antitrust laws.”⁷³ Kades commented: “Our view is that Deere’s argument contravenes the Supreme Court’s

⁶³ *Ibid.* *Eastman Kodak Co. v. Image Technical Services, Inc.*, 504 U.S. 451 (1992).

⁶⁴ *Ibid.*

⁶⁵ *Ibid.*

⁶⁶ Scarcella (2023b).

⁶⁷ *Ibid.*

⁶⁸ *In re: Deere & Company Repair Services Antitrust Litigation Statement of Interest of the United States* (14 February 2023).

⁶⁹ *Ibid.*, p. 2.

⁷⁰ *Ibid.*, p. 6.

⁷¹ *Ibid.*, p. 8.

⁷² Kades (2023).

⁷³ *Ibid.*

decision in *Kodak v. Image Technical Services*.⁷⁴ He noted: “Fortunately, Congress has given the Department of Justice authority to weigh in with courts to provide the department’s views – called a statement of interest.”⁷⁵

It would appear that the Biden administration will actively participate in the litigation, offering its views in respect of agricultural machinery, the right to repair, and competition policy.

2.5 Outcome

As the judge recognizes, this consolidated class action will be a “long and expensive process.”⁷⁶ The position of the farmers will be enhanced by the consolidation of the class action, and support from the U.S. Department of Justice. Nevertheless, John Deere is a very experienced, well-resourced, repeat-player in litigation. The outcome of the dispute between the farmers and John Deere is uncertain. The matter could be the subject of a judicial ruling and appeal, or could be settled in the interim. Nonetheless, the dispute – provisional as it still is – has crystallized key factual and doctrinal issues at stake in the conflict over the right to repair and agricultural machinery. The controversy has also raised further questions as to whether there should be further reform of competition laws in the United States. Economist Joseph Stiglitz has lamented: “Competition laws are eviscerated, incapable of responding to new technologies and firms’ new ways to get and exercise market power.”⁷⁷ It is submitted that the dispute should prompt an evaluation of the efficacy of American antitrust laws in the field of agriculture and farming.

3 Memorandum of Understanding

In an effort to defuse the litigation and conflict over the right to repair, John Deere entered into a memorandum of understanding in 2023 with the American Farm Bureau Federation.⁷⁸

The “purpose of this MOU is [to address repair] through a voluntary private sector commitment to outcomes rather than legislative or regulatory measures.”⁷⁹ The first purpose is to “continue to enhance the ability of Farmers to timely control the lawful operation and upkeep of Agricultural Equipment.”⁸⁰ The second purpose is to “assure the timely availability, on Fair and Reasonable terms, of Tools, Specialty Tools, Software and Documentation originating from Manufacturer, and Data from the operation of Agricultural Equipment originating from

⁷⁴ *Ibid.* *Eastman Kodak Co. v. Image Technical Services, Inc.*, 504 U.S. 451 (1992).

⁷⁵ *Ibid.*

⁷⁶ *Ibid.*

⁷⁷ Stiglitz (2024), p. 286.

⁷⁸ American Farm Bureau Federation (2023).

⁷⁹ *Ibid.*

⁸⁰ *Ibid.*

Manufacturer.”⁸¹ The third purpose is to “assure that no safety controls or protocols on Agricultural Equipment are compromised through the modification of protective measures installed for the benefit of Agricultural Equipment owners, operators and bystanders.”⁸² The fourth purpose is to “assure that the intellectual property of Manufacturer, including copyrighted software, is fully protected from illegal infringement through the modification of Embedded Software.”⁸³ The fifth purpose is to “assure that compliance with federal and state emissions control requirements is not compromised through changes to power ratings or other modification of control measures installed for the purpose of complying with the Clean Air Act and other environmental laws and regulations.”⁸⁴

Section III of the agreement maintains that the “AFBF agrees to encourage state Farm Bureau organizations to recognize the commitments made in this MOU and refrain from introducing, promoting, or supporting federal or state ‘Right to Repair’ legislation that imposes obligations beyond the commitments in this MOU.”⁸⁵ Section III provides: “In the event any state or federal legislation or regulation relating to issues covered by this MOU and/or ‘Right to Repair’ is enacted, each of AFBF and Manufacturer reserve the right, upon fifteen (15) days written notice, to withdraw from this MOU.”⁸⁶

There was significant media coverage of the 2023 Memorandum of Understanding between John Deere and the American Farm Bureau Federation – although the media did not necessarily show a good understanding of the limits of the agreement.⁸⁷

The National Agriculture Law Center notes that a number of other companies have also signed MOUs – including AGCO, CLAAS of America, CNH Industrial (including Case IH and New Holland), and Kubota.⁸⁸

There has been criticism of the John Deere Memorandum of Understanding as a poor substitute for law reform or antitrust enforcement. Professor Leland Glenna observed that “there’s a catch – the agreement isn’t legally binding, and, as part of the deal, the influential Farm Bureau promised not to support any federal or state right-to-repair legislation.”⁸⁹

PIRG Right to Repair Campaign Director Kevin O’Reilly commented:

If Deere truly provides farmers and independent mechanics with the same repair materials that its dealers have, then we would shout our praise from the rooftops. But the MOU contains limited enforcement mechanisms and the best aspects of this agreement could get lost in the legalese. Like Charlie Brown,

⁸¹ *Ibid.*

⁸² *Ibid.*

⁸³ *Ibid.*

⁸⁴ *Ibid.*

⁸⁵ *Ibid.*

⁸⁶ *Ibid.*

⁸⁷ Miller (2023), Huffstutter (2023), ABC Rural (2023), Roth (2023).

⁸⁸ The National Agriculture Law Center (2023).

⁸⁹ Glenna (2023).

farmers have lined up for the kick too many times to let Lucy pull the ball away again.⁹⁰

O'Reilly maintained that legislators “should move full steam ahead with their Right to Repair bills and not view this MOU as a sign that the problem is solved.”⁹¹ He argued that legislators “should continue pushing Right to Repair legislation until every farmer in every state with every brand of equipment can fix every problem with every tractor.”⁹²

Intellectual property expert Cory Doctorow highlighted that the Memorandum of Understanding failed to address the underlying intellectual property issues at stake with John Deere's tractors.⁹³ As such, he was of the view that the Memorandum of Understanding was only a shadow solution at best – because it did not deal with underlying problems.

In practice, there have been concerns that the Memorandum of Understanding has not delivered the repair freedoms, which farmers had been hoping for.⁹⁴

In October 2024, Senator Elizabeth Warren wrote to Deere & Company, accusing the company of undermining its own repair agreements by failing to grant its customers the right to repair their own agricultural equipment.⁹⁵ She argued: “Rather than uphold their end of the bargain, John Deere has provided impaired tools and inadequate disclosures.”⁹⁶ Warren commented: “In addition to costing farmers time and money, this refusal to provide meaningful right to repair likely indicates that John Deere may be violating the Clean Air Act by restricting repair of its products' emissions systems.”⁹⁷ Warren argues that “Deere's attempts to stave off right-to-repair reforms that would save American farmers \$4.2 billion per year are disgraceful.”⁹⁸

Warren's concerns are well-founded. It is submitted here that industry agreements are insufficient and inadequate means of protecting the right to repair, and a poor substitute for legislative action. The market itself will clearly not self-regulate the topic of repair effectively or fairly. Accordingly, there is a need for legislative action to enshrine the right to repair at a state level and at a federal level.

⁹⁰ PIRG (2023).

⁹¹ *Ibid.*

⁹² *Ibid.*

⁹³ Doctorow (2023).

⁹⁴ Bamberg (2024).

⁹⁵ Warren (2024).

⁹⁶ *Ibid.*

⁹⁷ *Ibid.*

⁹⁸ *Ibid.*

4 State Laws

In addition to taking litigation, United States farmers have been pushing for a right to repair at a legislative level.⁹⁹ In the United States, there has been a campaign to pass right to repair laws at a state level. There has been a singular instance of a successful campaign for a right to repair in agriculture in the State of Colorado.

4.1 Colorado

The State of Colorado has recognised a right to repair for farming equipment – the *Consumer Right to Repair Agricultural Equipment* HB23-1011 2023 (Colorado).¹⁰⁰

The summary of the legislation explained:

Starting January 1, 2024, the act requires a manufacturer to provide parts, embedded software, firmware, tools, or documentation, such as diagnostic, maintenance, or repair manuals, diagrams, or similar information (resources), to independent repair providers and owners of the manufacturer's agricultural equipment to allow an independent repair provider or owner to conduct diagnostic, maintenance, or repair services on the owner's agricultural equipment.¹⁰¹

The summary noted: "A manufacturer's failure to comply with the requirement to provide resources is a deceptive trade practice."¹⁰² The summary comments: "The act folds agricultural equipment into the existing consumer right-to-repair statutes and adds data to the list of resources that a manufacturer must provide to independent repair providers or owners."¹⁰³ However, "an independent repair provider or owner is not authorized to make any modifications to agricultural equipment that deactivates a safety notification system or brings the equipment out of compliance with safety or emissions laws or to engage in any conduct that would evade emissions, copyright, trademark, or patent laws."¹⁰⁴

The sponsor of the bill was Rep. Brianna Titone – a Democrat representative of Arvada, and a scientist by training.¹⁰⁵ She commented: "The livelihood of our farmers and ranchers are threatened by being forced to wait months and pay outrageous sums for equipment repairs because of our outdated repair laws."¹⁰⁶ Titone commented: "Right to repair gives farmers and ranchers the freedom to fix their own equipment in a timely manner, saving them money and getting them back

⁹⁹ Minter (2019), Betz (2020).

¹⁰⁰ *Consumer Right To Repair Agricultural Equipment* HB23-1011 <https://leg.colorado.gov/bills/hb23-1011>.

¹⁰¹ *Ibid.*

¹⁰² *Ibid.*

¹⁰³ *Ibid.*

¹⁰⁴ *Ibid.*

¹⁰⁵ Wilson (2023).

¹⁰⁶ Colorado House Democrats (2023).

to work faster.”¹⁰⁷ She maintained: “Agricultural equipment manufacturers shouldn’t be allowed to gatekeep parts, tools and information needed to repair farm equipment.”¹⁰⁸ In her view, “This bill puts an end to this predatory practice.”¹⁰⁹ (Titone was also the sponsor of a successfully passed right to repair bill for wheelchair users in Colorado).¹¹⁰ Titone was named Legislator of the Year by a grateful farmers union.¹¹¹ She noted: “This right to repair agricultural equipment law is a huge step forward in my ongoing legislative efforts to equip Coloradans with the tools to fix their own stuff when it breaks.”¹¹²

A key opponent of the bill was Rep. Richard Holtorf, R-Akron. He argued: “Those service departments might become distressed and it would cause the dealership to become distressed and then it goes away.”¹¹³ When something goes away, “it hardly ever comes back.”¹¹⁴

The legislation was approved by the Governor Democratic Jared Polis on 25 April 2023.¹¹⁵ Somewhat theatrically, Polis signed the bill, sitting in front of a Red Steiger 370 Tractor, during a ceremony outside the state capitol in Denver. Polis exclaimed that the bill was the “first in the nation!”¹¹⁶ Polis elaborated:

Colorado always likes to lead the way. Many farmers and ranchers face difficulty around exclusive arrangements where they have to get their repairs done by the dealer. ... This bill will save farmers and ranchers time and money by opening up additional options to repair farm equipment.¹¹⁷

The law became effective on the 1 January 2024.

The National Farmers Union was delighted by the Colorado bill.¹¹⁸ NFU President Rob Larew commented: “This is a huge win for farmers and ranchers in Colorado and across the country.”¹¹⁹ He observed: “NFU has been pushing on Right to Repair issues for years and seeing a bill like this cross the finish line is a testament to the persistence of our members and the need for this issue to be addressed nationally.”¹²⁰

¹⁰⁷ *Ibid.*

¹⁰⁸ *Ibid.*

¹⁰⁹ *Ibid.*

¹¹⁰ Colorado House Democrats (2022).

¹¹¹ Colorado House Democrats (2024).

¹¹² *Ibid.*

¹¹³ Metzger (2023).

¹¹⁴ *Ibid.*

¹¹⁵ Bedayn (2023b).

¹¹⁶ *Ibid.*

¹¹⁷ Metzger (2023).

¹¹⁸ National Farmers Union (2023).

¹¹⁹ *Ibid.*

¹²⁰ *Ibid.*

The Rocky Mountain Farmers Union was also a key advocate for the right to repair in Colorado.¹²¹ Chad Franke, president of Rocky Mountain Farmers Union, said: “This bill protects the rights of farmers and ranchers to choose how they repair their equipment and to have that work done in a timely manner.”¹²² He observed: “Lawmakers understood the immediate importance of this bill because of the farmers who showed up and told their stories of how right to repair makes sense.”¹²³

Franke commented: “Simply put, farmers and ranchers now have the freedom to choose how to repair the equipment that they own without the burdensome restrictions that could leave their equipment down at a critical time for their operation.”¹²⁴ He hoped that other states would follow suit: “While Colorado is the first state to accomplish this, we trust other states will follow our lead by allowing farmers and ranchers to work on their own equipment rather than have to wait for the manufacturers to pick and choose who gets service first and at profit from their exclusive hold on technology.”¹²⁵

Tyler Garrett – the head of government relations for the union – hoped that there would be federal action on the right to repair: “We need federal legislation that provides agricultural producers the ability to get their equipment fixed in a timely manner so that they can continue to provide the food, fiber and fuel to the people of Colorado, the United States and abroad.”¹²⁶

Right to repair advocates were pleased by the passage of the Colorado bill. Kevin O’Reilly, PIRG right-to-repair campaign director, commented: “With this bill, Colorado legislators are giving farmers the repair relief they deserve.”¹²⁷ He observed: “By passing a real right to repair, Colorado lawmakers are setting the standard for what right to repair can look like across the country.”¹²⁸ Willie Cade, a Repair.org board member and agricultural repair advocate, commented: “This legislation supports the great American tradition of competitive markets by expanding repair opportunities of equipment owners.”¹²⁹ Kyle Wiens, CEO of iFixit, commented: “The benefits of precision agriculture have come with software shackles that took away agency and local independence, imperilling the future of family farms.”¹³⁰ He was pleased that the Colorado “law goes a long way towards making ownership work the way we all expect.”¹³¹

¹²¹ Rocky Mountain Farmers Union (2023).

¹²² *Ibid.*

¹²³ *Ibid.*

¹²⁴ *Ibid.*

¹²⁵ *Ibid.*

¹²⁶ Toomer (2024).

¹²⁷ Nemetz (2023).

¹²⁸ *Ibid.*

¹²⁹ *Ibid.*

¹³⁰ *Ibid.*

¹³¹ *Ibid.*

For its part, John Deere was critical of the Colorado bill: “We feel strongly that the legislation in Colorado is unnecessary and will carry unintended consequences that negatively impact our customers.”¹³² The company maintained: “John Deere supports a customer’s decision to repair their own products, utilize an independent repair service or have repairs completed by an authorized dealer.”¹³³ The company argued: “John Deere additionally provides manuals, parts and diagnostic tools to facilitate maintenance and repairs.”¹³⁴

This article, though, would contend that the Colorado model is a positive, exemplary development – integrating a right to repair for agricultural machinery into a broader right to repair framework. The template of the legislative model should be picked up and adopted in other states in the United States.

4.2 Other States

Another dozen states in the United States are considering bills, which propose a right to repair in agriculture and farming.¹³⁵ Agricultural equipment owners have been seeking to delay or block or compromise such right to repair bills. The right to repair bill in West Virginia, for instance, has been undermined, after pressure from the tractor lobby.¹³⁶

Max Beal has called for the Nebraska Legislature to enact an Agricultural Right to Repair Act.¹³⁷ He observes that there was a bill introduced to the Nebraska Unicameral in 2021 (Legislative Bill 543) but that was not passed in the Spring 2022 legislative session. Beal argues: “The protections offered through such legislation are crucial because agricultural producers today are being deprived of the ability to make repairs to their equipment through protective software and End User License Agreements implemented by equipment manufacturers.”¹³⁸ He laments that “these restrictions on repair both directly and indirectly impact Nebraska’s economy.”¹³⁹ Beal maintains that “Nebraska has the authority to regulate the repair market, and by extension, has the power to mandate that manufacturers make repair tools available to consumers and independent repair providers.”¹⁴⁰ Beal contends: “Such an Act would curb the unfair trade practices being facilitated by manufacturers, thereby restoring agricultural producers’ right to repair their equipment.”¹⁴¹

¹³² *Ibid.*

¹³³ *Ibid.*

¹³⁴ *Ibid.*

¹³⁵ Bedayn (2023a).

¹³⁶ Culvyhouse (2024).

¹³⁷ Beal (2023).

¹³⁸ *Ibid.*, p. 246.

¹³⁹ *Ibid.*, p. 246.

¹⁴⁰ *Ibid.*, p. 264.

¹⁴¹ *Ibid.*, p. 278.

There has been a call for Kentucky to adopt legislation that secures farmers the right to fix their own equipment and machinery.¹⁴²

The State of Hawaii has been considering a right to repair bill – HB1884 and SB2496.¹⁴³ The House of Representatives received a range of testimony.¹⁴⁴ There was opposition from American Machinery – an agriculture and construction dealership, seemingly focused on John Deere equipment.¹⁴⁵

John Deere has backed lobbying groups involved in an anti-right to repair conference in 2021.¹⁴⁶ John Deere is a close ally of the Association of Equipment Manufacturers (AEM) – which has opposed the right to repair.¹⁴⁷ Such entities have helped set up the R2RSolutions initiative – which, in spite of its name, is an industry effort to address repair through voluntary agreements, without legislative intervention.¹⁴⁸

At present, there remains a lack of state protection for the right to repair in respect of agricultural machinery. The work of Lawrence Lessig has highlighted the ability of technology developers to forestall innovative, progressive law reform through political lobbying and donations.¹⁴⁹ The champions of the right to repair at a state level will need to build coalitions of supporters to follow the successful example of Colorado. Moreover, there will be a need to look to Federal law reform to try to provide comprehensive protection of the right to repair for agricultural machinery in the United States.

5 Federal Copyright Laws and the United States Copyright Office

The *Digital Millennium Copyright Act* 1998 (US) created new offences in respect of the circumvention of technological protection measures under copyright law.¹⁵⁰ There has been an ongoing constitutional challenge to the validity of the technological protection measures regime of the *Digital Millennium Copyright Act* 1998 (US). However, the latest court to hear that challenge has rejected arguments that the legislative power is beyond the power of the Congress, or in violation of the First Amendment.¹⁵¹

¹⁴² Miller (2024).

¹⁴³ *Relating to the Model State Right-to-Repair Law 2020* (Hawaii) HB1884 and SB2496 https://www.capitol.hawaii.gov/measure_indiv.aspx?billtype=HB&billnumber=1884&year=2020 and https://www.capitol.hawaii.gov/measure_indiv.aspx?billtype=SB&billnumber=2496&year=2020.

¹⁴⁴ Hawaii State House (2020).

¹⁴⁵ American Machinery (2020).

¹⁴⁶ Gault (2021).

¹⁴⁷ See for instance AEM (2020, 2021).

¹⁴⁸ R2R Solutions, <https://r2rsolutions.org/>.

¹⁴⁹ Lessig (2015, 2018, 2019).

¹⁵⁰ Litman (2001).

¹⁵¹ Green v. Department of Justice (2024) 2024 WL3628174 (US Court of Appeals for the District of Columbia Circuit).

In 2015, Deere and Company argued that Sec. 1201 of the *Digital Millennium Copyright Act* 1998 (US) gave it power to prevent purchasers of Deere Tractors from circumventing technological protection measures “for the purposes of lawful diagnosis and repair.”¹⁵² Deere and Company maintained: “The Copyright Office should not use its regulatory authority to facilitate myopic circumvention of TPMs, which would encourage the unauthorized copying and use of copyrighted software and related trade secrets of vehicle manufacturers and their suppliers.”¹⁵³

Ultimately, the United States Copyright Office was unconvinced by such arguments. In 2018, the United States Copyright Office recognised a right of repair as an exception to technological protection measures:

Multiple organizations petitioned to renew the exemption for computer programs that control motorized land vehicles, including farm equipment, for purposes of diagnosis, repair, and modification of the vehicle. The petitions demonstrated the continuing need and justification for the exemption to prevent owners of motorized land vehicles from being adversely impacted in their ability to diagnose, repair, and modify their vehicles as a result of TPMs that protect the copyrighted computer programs on the electronic control units (“ECUs”) that control the functioning of the vehicles Motor & Equipment Manufacturers Association, which during the sixth triennial rulemaking initially opposed any exemption that would impact the software and TPMs in vehicles, now supports the exemption as striking an appropriate balance between encouraging marketplace competition and innovation while mitigating the impact on safety, regulatory, and environmental compliance. The petitioners demonstrated personal knowledge and experience with regard to this exemption; each either represents or gathered information from individuals conducting repairs or businesses that manufacture, distribute, and sell motor vehicle parts, and perform vehicle service and repair.¹⁵⁴

There was much debate about the nature and the scope of such exceptions. The Acting Register recommended that, first, there be an exemption for “Computer programs that are contained in and control the functioning of a lawfully acquired motorized land vehicle such as a personal automobile, commercial vehicle or mechanized agricultural vehicle, except for programs accessed through a separate subscription service, when circumvention is a necessary step to allow the diagnosis, repair or lawful modification of a vehicle function, where such circumvention does not constitute a violation of applicable law, including without limitation regulations promulgated by the Department of Transportation or the Environmental Protection Agency, and is not accomplished for the purpose of gaining unauthorized access to other copyrighted works.”¹⁵⁵

In 2024, the United States Department of Justice and the Federal Trade Commission made a submission, arguing that there should be repair exemptions to

¹⁵² Deere and Company (2015).

¹⁵³ *Ibid.*, p. 2.

¹⁵⁴ United States Copyright Office (2018), pp. 17–18.

¹⁵⁵ *Ibid.*

the technological protection measures regime under the *Digital Millennium Copyright Act* 1998 (US).¹⁵⁶ The agencies argued: “In the Agencies’ view, facilitating greater freedom and choice in repairs will alleviate costly downtime due to broken equipment that results in a significant loss of revenue for businesses.”¹⁵⁷ The agencies cited the example of the breakdown of agricultural machinery in Nebraska discussed in testimony before the Nebraska legislature in 2016.

In 2024, Luke Hogg called on the United States Congress to reform copyright law and technological protection measures:

For nearly 25 years, Section 1201 has been hanging over the developers and distributors of tools that give users more control over the products they own. The ways in which John Deere and other corporations have used the copyright system is a glaring example of regulatory capture in action, highlighting the absurdity of a system where owning a product doesn’t necessarily convey the right to fully control it. There are certainly circumstances where the manufacturers are justified in protecting their products from tampering, but such cases should be handled through warranty nullification and contract law, not through exorbitant fines and lengthy prison sentences.¹⁵⁸

Hogg argued: “With the Farm Bill up for reauthorization, Congress is in a perfect position to reform Section 1201 of the DMCA, giving farmers, hackers, and the owners of all types of tech the freedom to repair.”¹⁵⁹

It is problematic that copyright law and technological protection measures have been deployed to frustrate the right to repair. Given that copyright law is focused upon encouraging cultural creations, it is a strange distortion of the regime for the laws to be used to protect agricultural machinery (which is far distant from the traditional subject matter of copyright law). The system of triennial administrative exceptions has been a piecemeal way to deal with the problem. There certainly does need to be holistic law reform of the *Digital Millennium Copyright Act* 1998 (US). However, it should be noted that copyright law reform itself will be insufficient. John Deere not only relied on copyright law and technological protection measures – but an array of other forms of intellectual property, including trade mark law, patent law, and trade secrets. The case study of John Deere should prompt a broader reconsideration of the need for right to repair defences across the various regimes of intellectual property.

6 Federal Law Reform Debate

There have been calls for a Federal Right to Repair law in respect of agricultural machinery. In a note, Gabriel Dominic Gomez comments about the struggle over the right to repair in agriculture and farming:

¹⁵⁶ United States Department of Justice and Federal Trade Commission (2024).

¹⁵⁷ *Ibid.*, p. 14.

¹⁵⁸ Hogg (2024).

¹⁵⁹ *Ibid.*

Right to repair is not just a farmer's crusade. It is fundamental to farmers to ensure they can optimize the food they grow. Furthermore, it is what consumers expect. The battle to win federal legislation for farmers or for any right to repair will not be easy. As farmers have learned, companies such as John Deere will work to stall legislation at every turn, even going so far as to strike a deal with no teeth in the eleventh hour of this battle, a deal that demanded no legislation.¹⁶⁰

United States competition experts have been concerned about the rise of monopolies in various sectors.

6.1 Agricultural Right to Repair Act

Senator Jon Tester – a Democrat from Montana – introduced the *Agricultural Right to Repair Act* to the 117th Congress (2021–2022).¹⁶¹ He discussed his motivation for the legislation:

I've been a farmer my whole life, and I've seen the unfair practices of equipment manufacturers make it harder and harder for folks to work on their tractors themselves – forcing them to go to an authorized mechanic and pay an arm and a leg for necessary repairs. Manufacturers have prevented producers from fixing their own machines in order to bolster corporate profits, and they've done it at the expense of family farmers and ranchers, who work hard every day to harvest the food that feeds families across the country. Farmers operate in tight windows and on tight margins, and they simply can't afford to waste time or money bringing their equipment to dealer authorized mechanics in the middle of a season. They need to be able to repair their own equipment, and this legislation will secure them that right.¹⁶²

Tester emphasized that the agricultural right to repair bill was “part of his continued effort to break up consolidation in the agriculture industry.”¹⁶³

The purpose of the bill was “to require original equipment manufacturers to make available certain documentation, parts, software, and tools with respect to electronics-enabled implements of agriculture, and for other purposes.”¹⁶⁴

Section 2 provided definitions of key terms, including: “authorized repair provider”, “commonly available”, “digital electronic equipment”, “documentation”, “electronics-enabled implement of agriculture”, “embedded software”, “fair and reasonable terms”, “firmware”, “independent repair provider”, “original equipment manufacturer”, “owner”, “part”, “software bill of material”, “tool”, and “trade secret.”

¹⁶⁰ Gomez (2024).

¹⁶¹ *Agricultural Right to Repair Act* S. 3549 – 117th Congress (2021–2022) <https://www.congress.gov/bill/117th-congress/senate-bill/3549>.

¹⁶² Tester (2022).

¹⁶³ *Ibid.*

¹⁶⁴ *Agricultural Right to Repair Act* S. 3549 – 117th Congress (2021–2022) <https://www.congress.gov/bill/117th-congress/senate-bill/3549>.

Section 3 laid down requirements for Original Equipment Manufacturers. Section 3(a) dealt with fair and reasonable terms. Section 3(b) dealt with disabling security functions. Section 3(c) concerned the interaction with copyright laws. In particular, it enabled diagnosis and repair of electronics enabled implementation of agriculture as an exemption to technological protection measures. It also supported interoperability, security testing, and non-infringing modifications to electronics enabled implement of agriculture. Section 3(d) sought to ensure that there was common availability of documentation, parts, software, and tools for repairers.

Section 4 focused on enforcement. Section 4(a) dealt with unfair or deceptive acts or practices. Section 4(b) concerned the powers of the Federal Trade Commission to take action under the Act. Section 5 deals with rulemaking.

Section 6 concerns limitations. Notably, Sec. 6(1) provides for the preservation of trade secrets of Original Equipment Manufacturers. Sections 6(2) and (3) says that the Act will not alter the terms of an agreement between an Original Equipment Manufacturer and an authorized repair provider. Section 6(4) says that the Act does not seek “to require an OEM to provide any part or equipment solely used in the development of their products.” Section 6(5) has further limitations regarding security; safety laws; emissions law; copyright laws; and other illegal modification activities.

The legislation received a number of endorsements from peak organisations in farming, and agriculture.¹⁶⁵ Montana Farmers Union President, Walter Schweitzer, commented: “Montana Farmers Union appreciates Senator Tester standing with family farms and ranches and demanding that equipment manufacturers allow us the right to repair our own equipment.”¹⁶⁶ Rob Larew, President of the National Farmers Union, commented: “Senator Tester’s bill would give farmers and independent mechanics the freedom to fix their equipment in a timely and cost-effective way.”¹⁶⁷ Sarah Carden, Policy Advocate for Farm Action, commented: “On behalf of the nation’s farmers and rural communities, we want to thank Senator Tester for introducing this legislation.”¹⁶⁸ Doug Crabtree, member of the Organic Trade Association Board of Directors, organic farmer and owner of Vilicus Farms in Northern Hill Country, Montana, observed: “This legislation would ensure we’re able to freely repair our farm machinery and make the most out of that investment.”¹⁶⁹

The legislation was also supported by organisations concerned with the right to repair, intellectual property, and consumer rights. Kevin O’Reilly, Right to Repair Campaign Director at the U.S. Public Interest Research Group, implored: “Farmers are asking for help – the Senate should pass the Agricultural Right to Repair Act to make it clear that they are listening.”¹⁷⁰ Kathleen Burke, Policy Counsel at Public Knowledge, observed: “This bill remedies one of the most outrageous abuses of

¹⁶⁵ Tester (2022).

¹⁶⁶ *Ibid.*

¹⁶⁷ *Ibid.*

¹⁶⁸ *Ibid.*

¹⁶⁹ *Ibid.*

¹⁷⁰ *Ibid.*

modern copyright law by ensuring that farmers will no longer have to ask the Copyright Office for permission to repair their equipment every three years.”¹⁷¹ George Slover, Senior Policy Counsel at Consumer Reports, said: “Consumer Reports strongly supports consumers having the right to repair their own equipment on their own terms, rather than having to always return to the manufacturer or its hand-picked technician.”¹⁷²

The legislation was also supported by organisations concerned with economic liberty, competition policy, and open markets. J. D. Scholten, Senior Advisor to the American Economic Liberties Project, commented: “With this legislation, Senator Tester is taking an important step towards revitalizing American agriculture and putting power back where it belongs: with American farmers, ranchers, and producers.”¹⁷³ Daniel Hanley, Senior Legal Analyst at the Open Markets Institute observed: “The Agricultural Right to Repair Act is the common-sense, long-overdue shield that farmers have been waiting for – it restores farmer access to the parts, tools, and software necessary to repair their equipment and do their jobs.”¹⁷⁴ The Open Markets Institute elaborated on its support in a separate press release: “The Agricultural Right to Repair Act is the common-sense, long-overdue shield that farmers have been waiting for – it restores farmer access to the parts, tools, and software necessary to repair their equipment and do their jobs.”¹⁷⁵

Rep. Marie Gluesenkamp Perez introduced a new version of the legislation to the 118th Congress – H.R. 5604 – the *Agricultural Right to Repair Act*.¹⁷⁶ Perez emphasized that the bill would establish “a comprehensive framework for the right to repair of agricultural equipment.”¹⁷⁷ The legislation was endorsed by the National Farmers Union, PIRG, and Farm Action Fund. William Biscoe, a farmer in Spotsylvania County, Virginia, said of the bill: “The Agricultural Right to Repair Act is a commonsense solution to help ensure farmers and ranchers are able to access the tools, technology and expertise needed to stay in the fields and produce the nation’s food supply.”¹⁷⁸

6.2 Other Bills

Oregon Representative Ron Wyden put forward a medical right to repair bill during the COVID-19 crisis.¹⁷⁹ He indicated that he was supportive of a general right to repair – as well as an agricultural right to repair – during that debate. Wyden and

¹⁷¹ *Ibid.*

¹⁷² *Ibid.*

¹⁷³ *Ibid.*

¹⁷⁴ *Ibid.*

¹⁷⁵ Open Markets Institute (2022).

¹⁷⁶ *Agricultural Right to Repair Act* H.R. 5604 – 118th Congress (2023–2024), <https://www.congress.gov/bill/118th-congress/house-bill/5604>.

¹⁷⁷ Perez 2024.

¹⁷⁸ *Ibid.*

¹⁷⁹ *Critical Medical Infrastructure Right-to-Repair Act* of 2020 (US) H.R. 7956 <https://www.congress.gov/bill/116th-congress/house-bill/7956> For commentary, see: Rimmer (2023).

Kullolli commented: “While this bill is focused on addressing the pandemic-induced emergency facing medical infrastructure, Americans should always have the right to repair the vehicles, tools, and devices they own.”¹⁸⁰ They hoped that there would be a right to repair in the field of agriculture: “If a farmer’s tractor breaks down, they should have the right to go to any qualified mechanic, instead of being locked-in to paying higher prices at dealerships.”¹⁸¹ Ron Wyden subsequently introduced the *Fair Repair Act* in 2024.¹⁸² The *Fair Repair Act* is co-sponsored by Senator Ben Ray Luján, D-N.M., and Congressman Joe Morelle, D-N.Y.

In 2023, Victoria Spartz (R-Ind.) introduced the *Farm Freedom to Repair Act*, to provide for the diagnosis, maintenance, and repair of certain digital electronic agricultural equipment.¹⁸³ This bill proposed an amendment to the copyright law provisions dealing with technological protection measures. Section 1201 of title 17, United States Code, would be amended by adding a new exception for the “Diagnosis, Maintenance, And Repair of Digital Electronic Agricultural Equipment.” The bill has been referred to the House Committee on the Judiciary.

More generally, Mondaire Jones put forward a *Freedom to Repair Act*.¹⁸⁴ Mondaire Jones was the U.S. Representative for New York’s 17th Congressional District from 2021 to 2023. The legislation would provide an exemption in the provisions dealing with copyright law and technological protection measures for the “Diagnosis, Maintenance, And Repair Of Digital Electronic Equipment.” However, as a result of redistricting, Mondaire Jones was forced to compete for the 10th Congressional District. He lost the primary to Dan Goldman.

The *National Defense Authorization Act* has been subject to proposed amendments from Rep. Marie Gluesenkamp Perez and Senator Warren to provide for the right to repair for the military.¹⁸⁵

6.3 Law Reform

As Lawrence Lessig has highlighted, intellectual property owners often campaign against progressive reforms to intellectual property law in the United States, seeking to delay or block the passage of such laws.¹⁸⁶ The United States political system has enabled economically powerful corporations to exercise significant political power, through political donations, lobbying, and interference.¹⁸⁷ In the context of the debate over the right to repair, there are certainly an array of technology developers

¹⁸⁰ Wyden and Kullolli (2020).

¹⁸¹ *Ibid.*

¹⁸² *Fair Repair Act* S. 4422, 118th Congress (2023–2024) <https://www.congress.gov/bill/118th-congress/senate-bill/4422>; and Wyden (2024).

¹⁸³ *Farm Freedom to Repair Act* H.R. 6879 – 118th Congress (2023–2024), <https://www.congress.gov/bill/118th-congress/house-bill/6879>.

¹⁸⁴ *Freedom to Repair Act* of 2022 – H.R. 6566 – 117th Congress (2021–2022) <https://www.congress.gov/bill/117th-congress/house-bill/6566/amendments>.

¹⁸⁵ National Defense Authorization Act (NDAA) – H.R. 8070/ S.4368 – 118th Congress (2023–2024) <https://www.congress.gov/bill/118th-congress/house-bill/8070/related-bills>; and Bowers (2024).

¹⁸⁶ Lessig (1999, 2001, 2004, 2008).

¹⁸⁷ Lessig (2015, 2018, 2019).

who are opposed to repair reforms, seeing them as an incursion into their intellectual property rights, and their profitable markets. The legislative champions of the right to repair will need to build a consensus across party lines to secure the passage of legislation through the House of the Representatives and the Senate, and obtain the approval of the United States President.

7 Executive Order and the Federal Trade Commission

A number of Democrat presidential contenders have voiced the need for an agricultural right to repair. As presidential candidate, Massachusetts Senator Elizabeth Warren vowed to legislate for a right to repair for farm machinery, during her Presidential campaign.¹⁸⁸ She called for a national right to repair law for farm equipment.¹⁸⁹ Warren contended: “Farmers should be able to repair their own equipment or choose between multiple repair shops.”¹⁹⁰ Likewise, Vermont Senator and Presidential Candidate Bernie Sanders was a vocal supporter for a right to repair for farmers.¹⁹¹ Another United States Democrat and presidential aspirant, Amy Klobuchar, has been particularly concerned about the need for antitrust reform to counter the rise of monopolies in the United States.¹⁹² Having won the Democrat nomination, and defeated Donald Trump in the election, President Joe Biden adopted some of the progressive policies of his Democrat contemporaries in respect of the right to repair.

In July 2021, President Joe Biden issued an executive order to promote competition.¹⁹³ This executive order highlighted competition problems in repair markets: “This order affirms that it is the policy of my Administration to enforce the antitrust laws to combat the excessive concentration of industry, the abuses of market power, and the harmful effects of monopoly and monopsony – especially as these issues arise in ... repair markets.”¹⁹⁴ The Executive Order encouraged the Federal Trade Commission to take action against “unfair anticompetitive restrictions on third-party repair or self-repair of items, such as the restrictions imposed by powerful manufacturers that prevent farmers from repairing their own equipment.”¹⁹⁵ The Federal Trade Commission voted to adopt this order and take enforcement action against repair restrictions.¹⁹⁶ AEM and John Deere responded to nervously to the Biden announcement.¹⁹⁷

¹⁸⁸ Warren (2019).

¹⁸⁹ Kelly (2019).

¹⁹⁰ *Ibid.*

¹⁹¹ Gault (2019).

¹⁹² Klobuchar (2021).

¹⁹³ The White House (2021a, 2021b).

¹⁹⁴ *Ibid.*

¹⁹⁵ *Ibid.*

¹⁹⁶ Federal Trade Commission (2021).

¹⁹⁷ AgWeb (2021).

The White House has subsequently convened a roundtable with federal and state officials, small business owners, and private sectors on the importance of the right to repair in 2023.¹⁹⁸ The readout included a focus on the right to repair in agriculture: “A farmer from Colorado and an independent repair shop owner from Minnesota explained how these legislative wins will save consumers and farmers money and create opportunities for small businesses.”¹⁹⁹

In 2022, a law firm Fairmark Partners LLP issued a complaint to the Federal Trade Commission against John Deere & Company on behalf of the National Farmers Union, many other farming unions around the country, as well as the U.S. Public Interest Research Group, the Digital Right to Repair Coalition, and iFixit.²⁰⁰ The submission requested that “the Commission investigate the repair restrictions Deere & Company (‘Deere’) has imposed that prevent farmers and ranchers from repairing their own Deere equipment or obtaining repairs from independent repair providers.”²⁰¹ The submission alleged that “Deere’s policy of withholding from its customers diagnostic software and other information necessary to repair the Deere equipment they own violates the Sherman Act and constitutes an unfair and deceptive trade practice, and they request that the Commission enjoin join Deere’s unlawful conduct.”²⁰²

Federal Trade Commission Chair Lina Khan has taken a keen interest in the right to repair.

In July 2024, Khan expressed concerns about the operation of repair restrictions in respect of agriculture: “As part of that work, we’ve been scrutinizing illegal repair restrictions, and we’ve heard a lot, including from farmers, about how, as we see more and more products become more digitized, manufacturers have more and more opportunities to restrict how you all can fix your own products.”²⁰³ She has observed that consultations with the farmers experiencing the effects of consolidation helps her and the Federal Trade Commission to understand the challenges faced in the field of agriculture: “In D.C., it’s really easy to just hear from the biggest companies or well connected lawyers, and it’s really important for us to be able to do our jobs as faithful public servants for us to actually hear from the public.”²⁰⁴

Colorado farmers and legislators such as State Rep. Brianna Titone have demanded that the Federal Trade Commission enforce right to repair regulations.²⁰⁵

The United States Congress has also been asking the Federal Trade Commission to take further action against repair restrictions. As the Chair of the Subcommittee on Competition, Antitrust, and Consumer Rights, Senator Amy Klobuchar – along with Senator Mike Lee – has written to Khan, asking her what steps the FTC is

¹⁹⁸ The White House (2023a, 2023b).

¹⁹⁹ *Ibid.*

²⁰⁰ Fairmark Partners LLP (2022).

²⁰¹ *Ibid.*

²⁰² *Ibid.*

²⁰³ Toomer (2024).

²⁰⁴ *Ibid.*

²⁰⁵ Colorado Newslines (2024).

taking to enforce existing laws against unlawful and anticompetitive repair restrictions.²⁰⁶ She noted: “These restrictions can increase repair and replacement costs for owners of products ranging from smartphones to farm equipment, prevent consumers and businesses from repairing equipment they own, impact consumers’ rights under existing law, limit opportunities for small and independent repair businesses, and create unnecessary harmful electronic waste.”²⁰⁷ The intent of the letter was to encourage the Federal Trade Commission to take further action to eliminate unlawful and anticompetitive repair restrictions.

Michael Carrier has argued that the Federal Trade Commission could use Sec. 5 of the *Federal Trade Commission Act* to strengthen the right to repair by taking action in respect of an “unfair method of competition.”²⁰⁸ He highlights that “the primary entity affected by anticompetitive repair restrictions is the user – [including] ... the farmer trying to get their tractor to work.”²⁰⁹ Carrier observes: “The combination of dealer consolidation and multiple manufacturers’ use of restrictions should lift John Deere’s market power above the threshold for liability.”²¹⁰ He comments John Deere’s justifications for its repair restrictions in terms of safety and intellectual property are questionable.²¹¹

In October 2024, it has been reported that the Federal Trade Commission is probing whether John Deere has violated Sec. 5 of the *Federal Trade Commission Act* through placing repair restrictions upon farmers.

In addition to the Federal Trade Commission, there has also been scrutiny of the role of the Environmental Protection Agency. In the 2024 case of *Cade v. United States Environmental Protection Agency*, there was a misconceived legal action over the right to repair against the United States Environmental Protection Agency.²¹² William Cade petitioned for a writ of mandamus to compel the United States Environmental Protection Agency and its Administrator, Michael S. Regan, in his official capacity (collectively “the EPA”), to enforce certain “right-to-repair” requirements against manufacturers of farm and other nonroad vehicles. Contreras J held: “Cade does not have standing to sue and neither statute provides a waiver of sovereign immunity as to the relief sought in Cade’s petition.”²¹³ The judge observed: “To the extent that Cade suffered an injury because of John Deere’s failure to provide “[e]mbedded [s]oftware repair capabilities” in connection with Cade’s purchase of “service software,” that injury cannot be traced to the challenged inaction of the EPA.”²¹⁴ The judge reflected: “Cade’s alleged injury

²⁰⁶ Klobuchar and Lee (2024).

²⁰⁷ *Ibid.*

²⁰⁸ Carrier (2022).

²⁰⁹ *Ibid.*, p. 1196.

²¹⁰ *Ibid.*, p. 1155.

²¹¹ *Ibid.*, p. 1155.

²¹² *Cade v. United States Environmental Protection Agency*, United States District Court, District of Columbia, 17 July 2024, Slip Copy, 2024 WL 3443470.

²¹³ *Ibid.*

²¹⁴ *Ibid.*

cannot be redressed by this Court; even if every right-to repair law and regulation were fully enforced by the EPA, Cade's injury would not be remedied."²¹⁵

The Biden Administration has certainly signaled that the right to repair is an important and significant matter of competition policy. The next United States President could and should take further action to realise the right to repair through executive orders and instructions for agency enforcement.

8 Conclusion

The monopolies in agricultural markets are raising profound issues in respect of intellectual property, farmers' rights, competition policy, and sustainable development.

The current United States law on the right to repair and agricultural machinery remains fragmented and inchoate. There has been a consolidated class action by farmers against John Deere, alleging breaches of antitrust law. At a preliminary stage thus far, the judge has allowed the dispute to proceed. The dispute will take a long while to work its way through the courts. The Memorandum of Understanding promoted by the American Farm Bureau Federation and John Deere will not necessarily resolve the larger concerns about the right to repair. The state-based movement to recognize an agricultural right to repair has found favour in Colorado, with the passage of a new regime. The company John Deere has relied upon a combination of intellectual property rights to maintain a tight control of its agricultural technologies. At a Federal level, the United States Copyright Office has permitted an exception under copyright law and technological protection measures to allow for the repair of agricultural machinery. However, the exception is a narrow and circumscribed one. There have been a number of Federal congressional bills put forward to provide for an agricultural right of repair – the most notable being Senator Jon Tester's *Agricultural Right to Repair Act*. There needs to be further political consensus to secure the passage of an agricultural right to repair in the United States Congress. At an executive level, President Joe Biden has issued an executive order to promote competition – mentioning in particular the right to repair in agricultural markets. The Federal Trade Commission has been starting to take enforcement action and policy consultation to fully realise the right to repair. Lina Khan has taken a personal interest in the top of the agricultural right to repair. While this wide-ranging action is promising, there is a need to consolidate and crystallize such initiatives. There needs to be concerted action to recognize an agricultural right to repair in the United States – which addresses comprehensively questions of intellectual property, competition policy, and farmers' rights.

In intellectual property, there has long been a discussion of the importance of farmers' rights, particularly in the context of plant breeder's rights and patent law.²¹⁶ There has also been discussion of farmers' rights in the context of the International Treaty on Plant Genetic Resources.²¹⁷ Special rapporteur on the right to food Michael Fakhri has argued that there should be a recognition of farmers' and

²¹⁵ *Ibid.*

²¹⁶ Smith and Bragdon (2016).

²¹⁷ Prip (2024).

Indigenous peoples' rights to seed.²¹⁸ There has been debate about modernizing the concept of farmers' rights.²¹⁹ One of the components of a broadly conceived set of farmers' rights is the right to repair. In her book, *Break 'em Up*, Professor Zephyr Teachout discusses the concentration of economic and political power in agricultural markets.²²⁰ She highlights the issue of the right to repair: "Even on his own tractor, a farmer has no freedom: John Deere contracts require that tractors be serviced by affiliates of the company."²²¹ She hopes: "With a grassroots, anti-monopoly movement: Farmers can reclaim the right to fix their own tractors."²²² The broad recognition for a right to repair for farmers will provide for exceptions and limitations to intellectual property rights.

The case study of John Deere and the right to repair has also highlighted larger questions about the inter-relationship between intellectual property and competition policy. As Professor Jill McKeough has noted, "Providing consumers with better, more affordable products, promoting higher living standards and expanding the choices, income and benefits to society are the aims of both competition policy and intellectual property laws, and the two are largely complementary."²²³ However, she highlights "tensions between these regimes that arise because there is a potentially anti-competitive cost to granting intellectual property rights."²²⁴ There is a need for intellectual property law, policy, and practice to incorporate competition policy into its internal doctrines. Moreover, intellectual property law, policy, and practice should be subject to external oversight by competition regulators. Indeed, competition regulators have a key role to play in ensuring fair competition in the aftermarkets for repair of agricultural machinery. The Federal Trade Commission in the United States has certainly expressed an interest in the topic of the right to repair – but at present has been tentative about taking enforcement action. Likewise, in Australia, the Australian Competition and Consumer Commission has held policy inquiries into the topic of the right to repair in respect of agricultural machinery. However, as noted by the Productivity Commission, the Australian Competition and Consumer Commission could and should have played a much more active role in taking enforcement action in respect of repair restrictions. Competition regulators have tended to be more active in the European Union in new technology markets.²²⁵ There is a growing scholarship on the tensions between intellectual property and competition law.²²⁶ The conflicts also perhaps highlight problems of monopolies forming in highly concentrated agricultural markets. From an international perspective, there is a need for nation

²¹⁸ Fakhri (2021).

²¹⁹ Adhikari et al. (2021).

²²⁰ Teachout (2020).

²²¹ *Ibid.*, p. 26.

²²² *Ibid.*, p. 7.

²²³ McKeough (2003), p. 290.

²²⁴ *Ibid.*, p. 290.

²²⁵ Imarhiagbe (2022).

²²⁶ Drex1 (2010), Carrier (2011), Kaesberg (2012), Treacy (2024).

states to make use of competition flexibilities available under international intellectual property agreements – such as the TRIPS Agreement 1994.

Although this article has focused on the United States, it should be noted that the United States is not necessarily an isolated case or an exceptional case in respect of the debate over John Deere. There are a number of other jurisdictions which have been actively exploring recognition of the agricultural right to repair. In Australia, the Australian Competition and Consumer Commission has undertaken a number of studies of the aftermarkets for repair of motor vehicles, and agricultural machinery.²²⁷ The Productivity Commission has undertaken an inquiry into the right to repair – recommending in particular action on an agricultural right to repair.²²⁸ While sympathetic to the call for a right to repair, the Albanese Government has been slow to respond to the recommendations of the Productivity Commission.²²⁹ In Canada, Natasha Tusikov has called for the recognition of an agricultural right to repair.²³⁰ She comments: “Choice in repair is also important for Canadian farmers, given the cost and difficulty of access to manufacturer-authorized repairers, especially during brief planting or harvest seasons.”²³¹ In the European Union, there has historically been competition concerns about John Deere.²³² There has been recognition of the need to share repair information for agricultural machinery.²³³ John Deere and allied groups such as CEMA have lobbied against the recognition of a full agricultural right to repair.²³⁴ The European Parliament and the European Commission have passed a directive focusing on the right to repair – but at present it focuses on consumer goods,²³⁵ but not agricultural machinery.²³⁶ There have been calls for a further expansion of the right to repair in the European Union.²³⁷ A full discussion of the comparative debate over the agricultural right to repair is beyond the scope of this paper – but nonetheless further work should be

²²⁷ Australian Competition and Consumer Commission (2017), Australian Competition and Consumer Commission (2018), Australian Competition and Consumer Commission (2020); and Australian Competition and Consumer Commission (2021).

²²⁸ Productivity Commission (2021).

²²⁹ ABC Rural (2023), Powell et al. (2023), Leigh (2024).

²³⁰ Tusikov (2024).

²³¹ *Ibid.*

²³² *John Deere & Co. v. N.V.. Cofabel*, 14 December 1984 O.J.L. 35; and *John Deere Ltd v. Commission of the European Communities*, Judgment of the Court of First Instance (Second Chamber) of 27 October 1994. European Court Reports 1994 II-00957.

²³³ Consolidated text: Regulation (EU) No 167/2013 of the European Parliament and of the Council of 5 February 2013 on the Approval and Market Surveillance of Agricultural and Forestry Vehicles <http://data.europa.eu/eli/reg/2013/167/2019-04-18>.

²³⁴ Dryancour (2011), CEMA (2018), CEMA (2024).

²³⁵ Common Rules Promoting the Repair of Goods. European Parliament Legislative Resolution of 23 April 2024 on the Proposal for a Directive of the European Parliament and of the Council on Common Rules promoting the Repair of Goods and amending Regulation (EU) 2017/2394, Directives (EU) 2019/771 and (EU) 2020/1828 (COM(2023)0155 – C9-0117/2023 – 2023/0083(COD)). Date: 23-04-2024 Reference :P9_TA(2024)0308 and European Parliament (2024).

²³⁶ Kennedy (2024).

²³⁷ European Environmental Bureau (2023), European Law Institute (2023), Generation Climate Europe (2023); and Right to Repair Europe (2024).

done exploring the litigation and policy developments over the agricultural right to repair in other jurisdictions.

This article on the right to repair in the field of agriculture raises larger questions about the need for an international framework for the right to repair. The recognition of a broad right to repair in the field of agriculture will help promote the United Nations Sustainable Development Goals. There has been much discussion about the need to realign intellectual property law, practice, and policy with sustainable development.²³⁸ Indeed, the World Intellectual Property Organization highlighted “IP and the SDGs” as a key theme for World Intellectual Property Day in 2024.²³⁹ The adoption of a broad right to repair could an early step in the transformation of intellectual property to promote sustainable development.²⁴⁰ The combination of intellectual property reform and effective competition policy could play a role in addressing the climate crisis.²⁴¹ The recognition of a right to repair would promote responsible consumption and production (SDG 12), and the development of a circular economy.²⁴² Moreover, the adoption of a right to repair would support life on the land (SDG 15), and food security (SDG 2). The adoption of a right to repair would also promote local innovation (SDG 9), and address inequalities between farmers and the corporation section (SDG 10). It would be helpful and useful to develop a broader international framework in the form of an agreement, a treaty, a convention, or a declaration on the right to repair.

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²³⁸ Rognstad and Orstavik (2021).

²³⁹ World Intellectual Property Organization (2024).

²⁴⁰ Ozturkcan (2023).

²⁴¹ Drexl (2024).

²⁴² Pihlajarinne (2021).

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